

Nikolaos Vynios — M2 Internship (May–September 2026)

Supervisors: John Samuel, Gilles Gesquière — IA.rbre project

From image to volume: generalizing the IA.rbre paradigm towards a voxelized representation for volumetric modeling of territories

My internship subject

Title

“From image to volume: generalizing the IA.rbre paradigm towards a voxelized representation for volumetric modeling of territories”

Objective

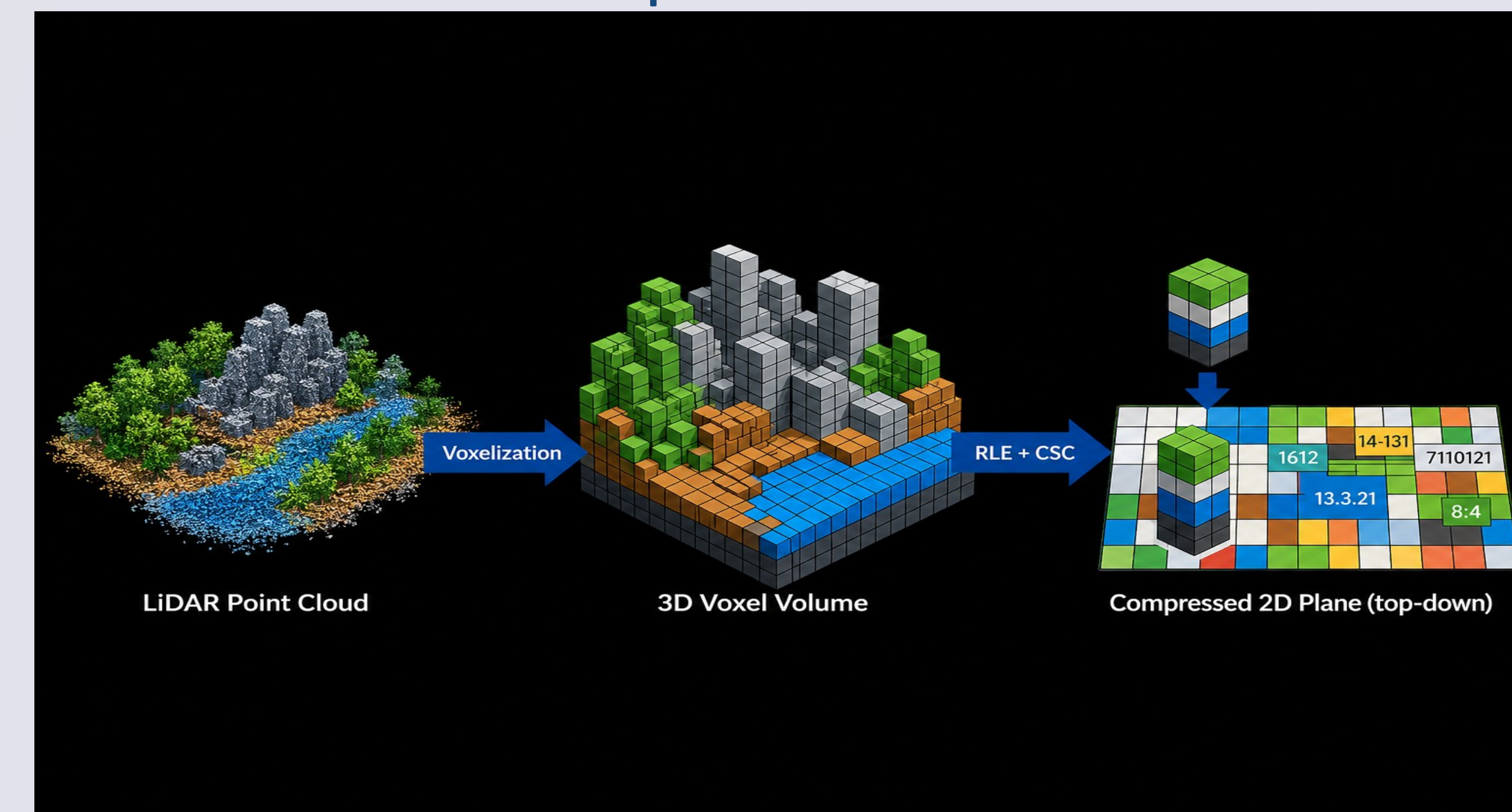
- Move from LiDAR to a 3D volumetric representation
- Compress 3D representation on a 2D plane

Target technical applications

- Better estimation of cast shadow, thermal screening, 3D plantability
- Integration with VCity 3D visualization tools (UD-Viz, UD-SV)

Internship duration

- M2 internship · May 2026 – September 2026 (4 months)



State of the Art: Voxelized Representation

1. Point Cloud Encoding Strategies

[1] Huang et al. (2023) — Voxel-Pillar Fusion (VPF): hybrid network combining 3D voxel and 2D pillar sparse convolutions with bidirectional Sparse Fusion Layer. AAAI.

[2] Peytavie (2010) — PhD thesis: procedural world generation via unified volumetric voxel terrain model. Supports caves, overhangs, tunnels, bridges. Univ. Lyon 1.

2. Voxelisation Algorithms & Data Structures

[3] Gomes et al. — "Voxelisation Algorithms and Data Structures: A Review" (PMC8707769). Surveys regular grids, octrees, k-d trees; surface vs. solid voxelization, memory trade-offs.

[4] Schnabel et al. (2011) — Incremental segmentation of massive LiDAR point clouds using an octree-structured voxel space. *Computer Graphics Forum*, Wiley.

3. Classification & Segmentation

[5] Lu et al. (2023) — Voxel classification of aerial urban LiDAR: voxel attributes (planarity, sphericity, omnivariance) with Random Forest for buildings, vegetation, ground.

[6] Poux & Billen (2019) — Voxel-based semantic segmentation: shape features (SF1) and 3D structural connectivity (SF2) vs. deep learning. F1 >85% on S3DIS. *IJGI*.

4. 3D Data Integration

[7] ISPRS Annals (2024) — "3D Data Integration in the Voxel Domain". Unified voxel representation for heterogeneous 3D geospatial data: point clouds, meshes, CityGML.

[1] Voxel-Pillar Fusion Network (Huang et al., 2023)

Voxel or Pillar: Exploring Efficient Point Cloud Representation for 3D Object Detection

Huang Y., Zhou S., Zhang J., Dong J., Zheng N.

Institute of Artificial Intelligence and Robotics, Xi'an Jiaotong University



西安交通大学
XI'AN JIAOTONG UNIVERSITY

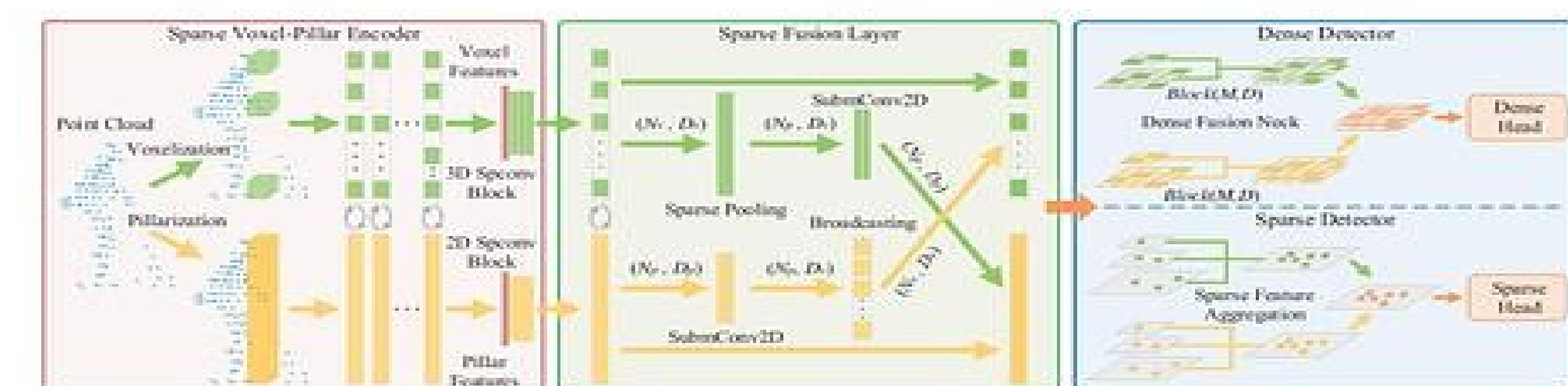


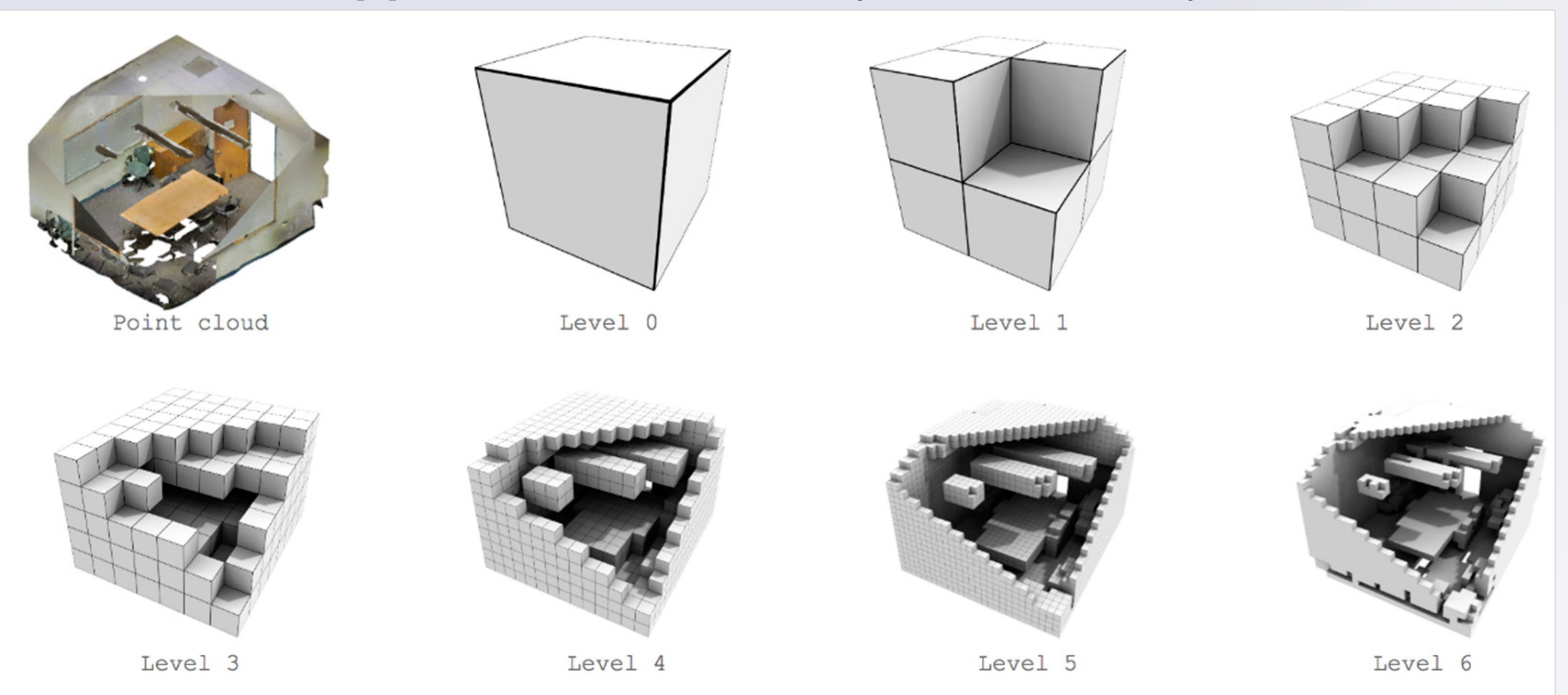
Figure 1. The framework of Voxel-Pillar Fusion (VPF) Network. We extract sparse voxel and pillar features via a dual-branch encoder. The Sparse Fusion Layer is presented for bidirectional interaction between sparse features. Together, these components form a hybrid backbone capable of integrating with both dense and sparse detectors.

Distinctions of the Voxel and Pillar Representation

grid-based methods quantize the point cloud into regular voxels or pillars and generate bounding boxes in 2D Bird's Eye View (BEV), achieving

Detection Framework. Serving as a fully sparse hybrid representation backbone, our method can be seamlessly incorporated into different types of detectors. We introduce VPF_{1c} and VPF_{1s} as variants of dense and sparse detectors, respectively. For VPF_{1c}, sparse features are first

[6] Octree-Based Voxel Levels (Poux & Billen, 2019)



Progress made

Literature review and onboarding

- ▶ Literature review: urban vegetation segmentation (LiDAR, deep learning, fusion)
- ▶ FLAIR-HUB, Myria3D, COSIA syntheses, available in github.com/VCityTeam/UD-IArbre-Research
- ▶ Study of existing pipeline: code et documentation sur GitHub du projet IA.rbre

Technical setup

- ▶ Testing of the vegetalisation/pipeline on Lyon Tiles

First reflections

- ▶ Preliminary definition of a candidate voxelized representation
- ▶ Identification of key steps in the 3D pipeline to develop

Planning

The approach

- ▶ LiDAR → voxel grid → store each vertical column as a compressed list
- ▶ RLE: a 30 m column (150 slots) shrinks to 3–5 entries — only store occupied voxels
- ▶ CSC: each column stores (row_index, value) pairs for non-empty cells only

Getting it working

- ▶ Read LiDAR files, split into columns, produce first compressed representation
- ▶ Validate: column count, entries per column, visualization

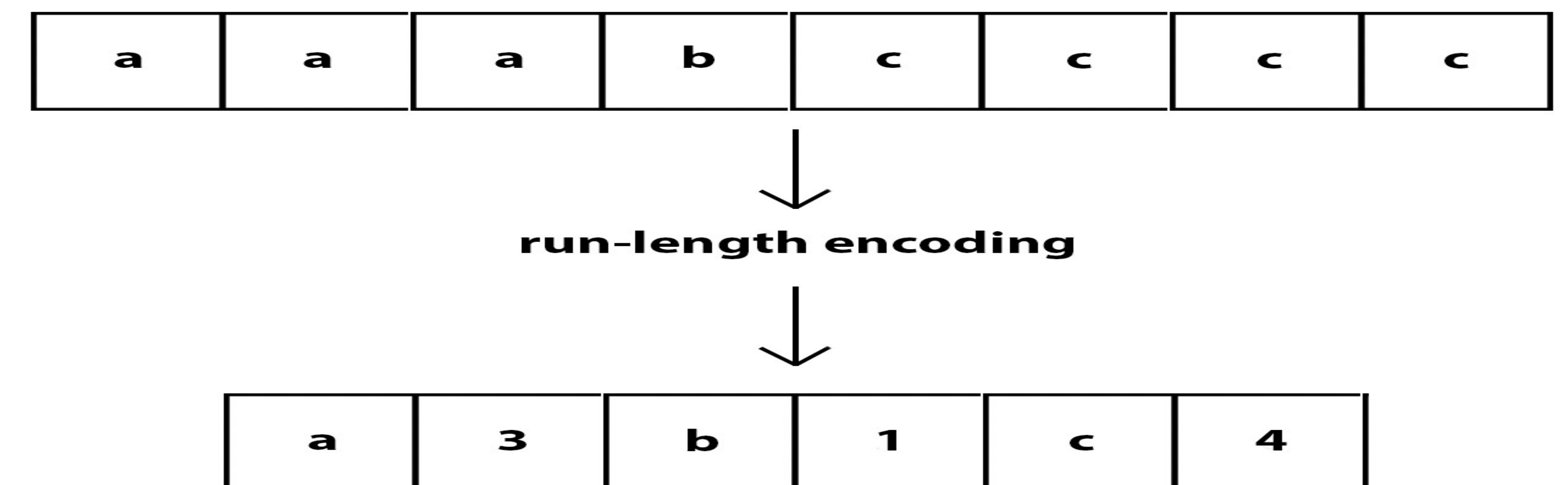
Enrichment

- ▶ Attach color (orthophoto) and vegetation labels (FLAIR) per column
- ▶ Test query efficiency: random access, range queries

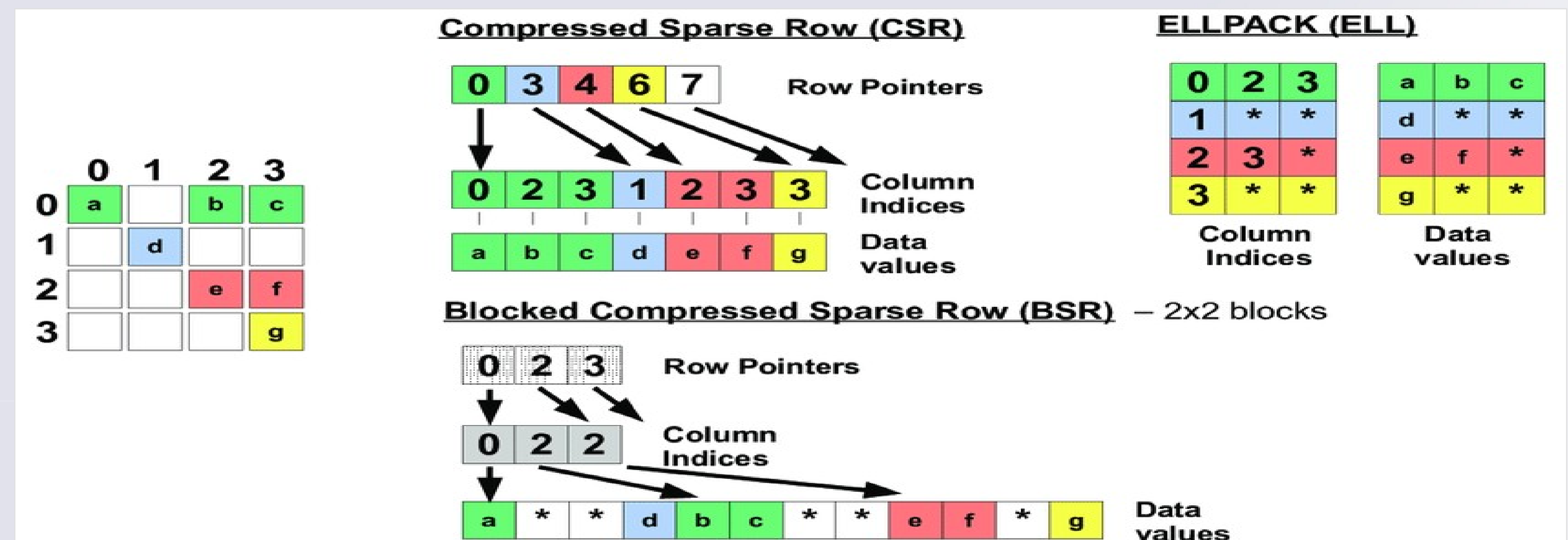
Evaluation and delivery

- ▶ Compare against existing 2D pipeline output
- ▶ Documentation, internship report, final presentation

Run-Length Encoding (RLE) — Voxel Column Compression



Compressed Sparse Column (CSC) — Sparse Matrix Storage



Refs: [1] Golomb (1966), "Run-Length Encodings", *IEEE TIT*. [2] Saad (2003), *Iterative Methods for Sparse Linear Systems*, SIAM (CSC/CSR). [3] Voxel Compression with RLE & space-filling curves ([eisenwave.github.io](https://github.com/eisenwave)).

Conclusion

Assessment after 3 weeks

- ▶ Good understanding of the context: IA.rbre, data, technical environment
- ▶ Good understanding of the existing 2D pipeline (LiDAR HD + FLAIR-HUB)
- ▶ Start of technical exploration towards 3D representation

Sources of Inspiration

- ▶ **Peytavie (2010)** — PhD thesis: unified *volumetric voxel terrain model* for representing different materials, enabling caves, overhangs, tunnels, and bridges.

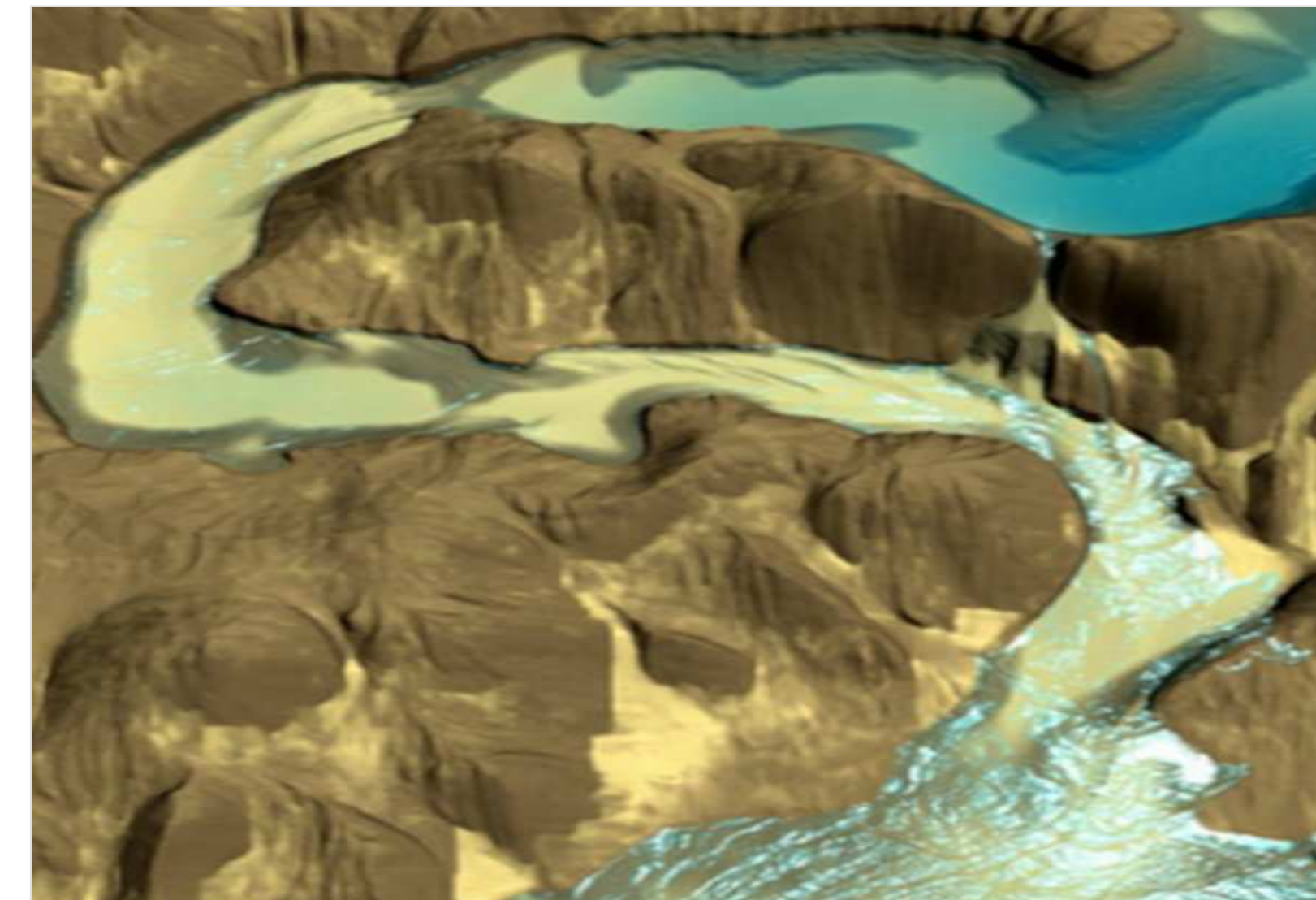
[HAL tel-00841373](#) — Univ. Claude Bernard Lyon 1, LIRIS

Open questions — your feedback welcome

- ▶ Optimal voxel resolution for Lyon data
- ▶ 2D↔3D transition strategies
- ▶ Efficient query mechanisms for RLE+CSC voxel columns

Thank you for your attention!

Volumetric terrain (canyons)



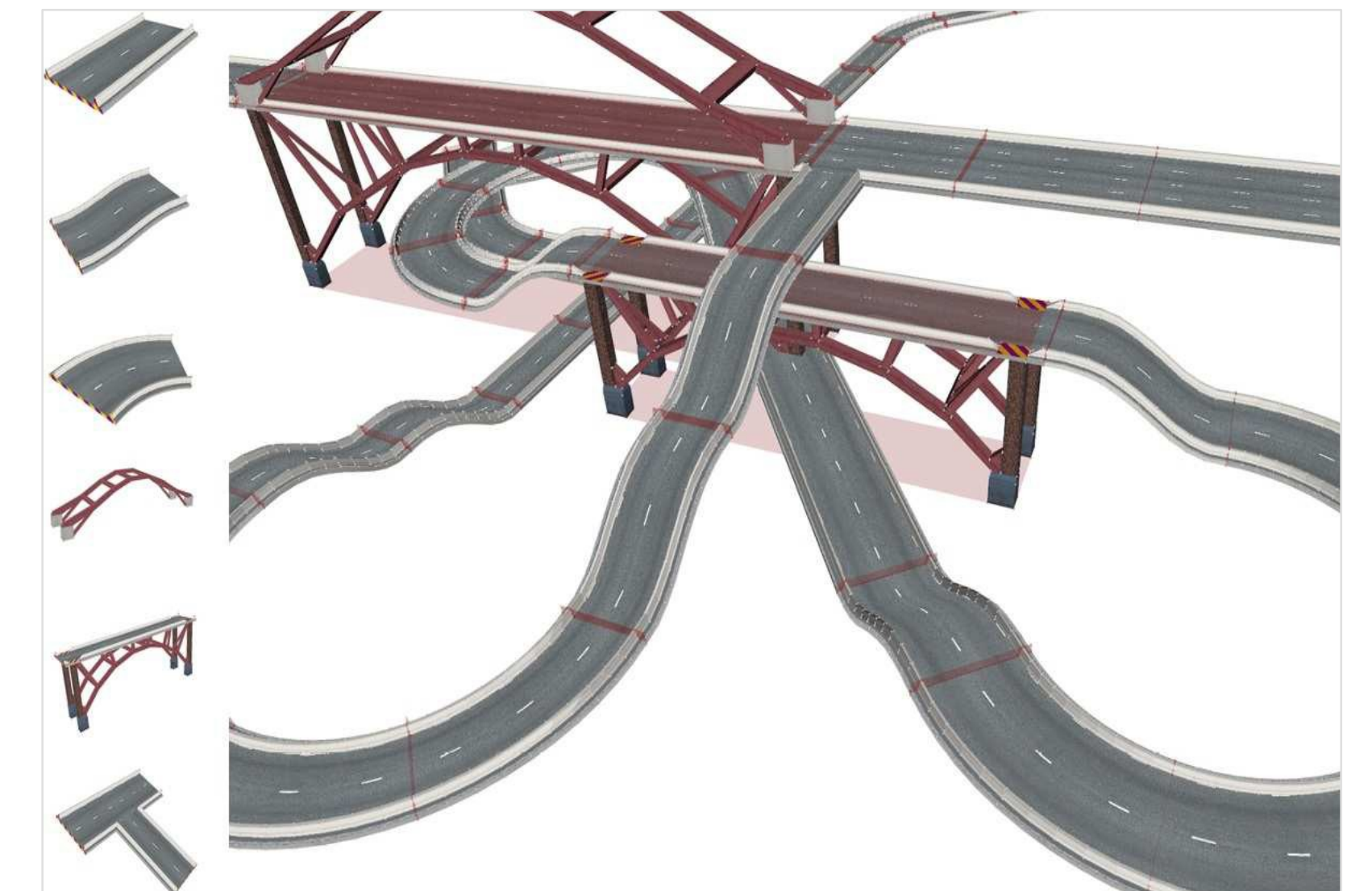
Overhangs & arches



Caves & tunnels



Bridges & infrastructure



Peytavie, A. (2010). *Génération procédurale de monde*. PhD thesis, Université Claude Bernard — Lyon I, LIRIS. [HAL tel-00841373](#).